Rahul Shewatkar

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Portfolio: https://rshewatkar.github.io/portfolio-website/

GitHub: https://github.com/rshewatkar

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Profile

AI/ML engineer specializing in logistics-domain AI applications. Built production RAG systems, LLM-powered document processing pipelines, and agentic workflows using Python, LangChain, FastAPI, and Docker. Combining 4+ years of logistics operations expertise with hands-on AI deployment experience to build intelligent supply chain automation.

EDUCATION

Shree H.V.P. M's COET Amravati, Amravati University | Jun 2013 – May 2016

Bachelor of Engineering in Electronics and Telecommunication

P.R. Pote Patil Institute Polytechnic, Amravati | Jun 2010 – May 2013

Diploma in Engineering

SKILLS

Programming Languages : Python, SQL (Basic)

AI / Machine Learning : Pytorch, Scikit-learn, XGBoost, H2O AutoML, TensorFlow, Feature Engineering, Model Evaluation, Predictive Analytics

Generative AI & LLM Engineering : LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Hugging Face Transformers Sentence Transformers, Agentic AI, Prompt Engineering, Supply Chain AI Applications

Data Engineering & Visualization : Pandas, NumPy, Exploratory Data Analysis (EDA), Data Preprocessing, Matplotlib, Seaborn

MLOps & Deployment : MLflow, Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, FastAPI, Streamlit, Render, Hugging Face Spaces

Vector Databases & Retrieval : ChromaDB, Qdrant , Vector Search, Semantic Search, Embeddings

Knowledge Graph : Neo4j

Cloud Platforms : AWS (Familiarity), Azure ML (Exposure), GCP (Exposure; No Production Experience)

Observability : Prometheus, Grafana , CloudWatch

Developer Tools : Git, GitHub, Jupyter Notebook, VS Code

Domain Expertise : Logistics AI, Supply Chain AI, Freight Analytics, Shipment Delay Prediction, Demand Analysis

EXPERIENCE

Codec Technologies Pvt. Ltd. **Mumbai, India**

Data Science Intern **Aug/2025 – Nov/2025**

- Worked on end-to-end machine learning pipelines, from data preprocessing and feature engineering to model training and evaluation.
- Built and tested supervised learning models (regression & classification) using Python, focusing on performance, accuracy, and business relevance.
- Conducted exploratory data analysis (EDA) to identify patterns, correlations, and data quality issues that informed model design.
- Applied feature selection and transformation techniques to improve model stability and predictive performance.
- Documented workflows, results, and insights to support reproducibility and stakeholder understanding.

Unified Mentor Pvt. Ltd. **Gurugram, India**

Machine Learning Engineer Intern **Jan/2025 – Jul/2025**

- Contributed to price prediction and classification pipelines; performed EDA using Pandas and Seaborn to identify key correlations across structured datasets of 10K+ records.
- Contributed to data cleaning, preprocessing, and feature engineering tasks using Python (Pandas, NumPy).
- Supported exploratory data analysis and visual storytelling using Matplotlib and Seaborn.
- Assisted in model training and evaluation for real-world use cases such as price prediction and classification.
- Documented findings, created dashboards, and presented results to mentors and project stakeholders.

Shipco IT Pvt. Ltd. **Pune, India**

Export Documentation Executive **Sep/2019 – Feb/2023**

- Prepared and managed all export documentation, including invoices, packing lists, and bills of lading, ensuring compliance with international regulations.
- Coordinated shipments to international customers across the US, Europe, and Asia, achieving on-time delivery rate.
- Led the migration of the Manifest process for major ports like Keelung, Rotterdam, and Shanghai, improving operational accuracy.
- Handled Master Bill of Lading (MBL) queries and collaborated with international offices to resolve discrepancies promptly.

WNS Global Services Pvt. Ltd. **Pune, India**

Associate Operations (Air Freight) **Mar 2018 – Apr 2019**

- Managed customer onboarding and generated credentials for air freight tools, ensuring smooth system access.
- Compiled monthly performance reports and presented insights to clients, driving data-driven decision-making.
- Calculated and allocated reward points based on revenue data, enhancing customer engagement and retention.

GTL Pvt. Ltd. (Micro-Internship) **Pune, India**

Engineer **Feb/2018**

- Monitored and resolved network issues by collaborating with escalation and incident management teams.
- Delivered satisfactory services by managing customer-end devices and troubleshooting network problems efficiently.

PROJECTS

Supply Chain Intelligence Platform

Python, Neo4j, NetworkX, FastAPI, Streamlit, Qdrant, Sentence Transformers, LangChain, OpenAI-compatible LLMs

Developed an AI-powered supply-chain intelligence platform that analyses company relationships, supplier dependencies, and supply-chain risks using knowledge graphs and Retrieval-Augmented Generation (RAG).

- Knowledge Graph Engineering: Built a Neo4j knowledge graph covering 10 companies, representing supply-chain entities and relationships to analyses supplier networks and dependencies.
- Data Processing & Semantic Search: Developed document-processing and embedding pipelines using Sentence Transformers and Qdrant, indexing 87,000+ semantic chunks for contextual information retrieval.
- Graph Analytics & Risk Assessment: Implemented graph-based analytics using NetworkX and Neo4j, including degree, betweenness, closeness centrality, community detection, and weighted entity risk scoring.
- AI-Powered RAG System: Developed a retrieval and question-answering pipeline that combines semantic search, contextual prompts, and LLM-generated responses for supply-chain intelligence.

- REST API Development: Built modular FastAPI endpoints for graph exploration, supplier analysis, risk analytics, and AI-assisted supply-chain queries.
- Interactive Dashboard: Developed a Streamlit dashboard with company intelligence, graph exploration, risk monitoring, analytics, and AI assistant pages.
- **GitHub:** https://github.com/rshewatkar/Supply-Chain-Intelligence-Platform

RAGOps-AI – Production-Ready RAG & MLOps Pipeline

Tech Stack: FastAPI, Streamlit, MLflow, Docker, GitHub Actions, ChromaDB, HuggingFace

Designed an agentic document Q&A pipeline using LangChain retrieval chains with tool-calling capabilities, enabling multi- step reasoning over resume content with context-aware memory across conversation turns.

- Developed a production-style AI Resume Assistant using Retrieval-Augmented Generation (RAG) architecture.
- Built REST APIs with FastAPI and interactive frontend with Streamlit.
- Designed vector database pipeline using ChromaDB and sentence-transformer embeddings.
- Tracked experiments and retrieval metrics with MLflow for prompt, chunk-size, k-value, and relevance evaluation.
- Implemented Docker-based microservice deployment and CI/CD automation using GitHub Actions.
- Deployed multi-service architecture on Render with separate API, UI, and MLflow tracking services.
- **Live app:** https://ragops-streamlit-ui.onrender.com
- **GitHub:** https://github.com/rshewatkar/ragops-ai

Shipment Delay Predictor

Tech Stack: Scikit-learn, XGBoost, Streamlit, Docker, GitHub Actions, HuggingFace

- Built an intelligent shipment delay prediction system for logistics and courier operations — automating risk assessment before warehouse dispatch using XGBoost with F1=71.51% and AUC-ROC=75.29%, directly reducing manual review effort for operations teams.
- Applies domain expertise from 4+ years in international freight operations (Shipco IT, WNS Global) to engineer logistics-specific features: discount tiers, weight bands, and customer escalation signals.
- Performed EDA on 10,999 shipments, uncovering key delay drivers: high discounts, 2–4kg weight range, and 4+ customer care calls. Engineered 3 new features from these insights
- Built an interactive Streamlit dashboard for operations teams to input shipment details and get instant delay risk predictions with actionable recommendations
- Deployed as a live Streamlit app on Hugging Face Spaces, allowing real-time delay risk assessment
- **Live app:** https://rshewatkar-shipment-delay-predictor.hf.space/
- **GitHub:** https://github.com/rshewatkar/shipment-delay-predictor

Ship CO_2 Emission Predictor

Tech Stack: Scikit-learn, Streamlit, GitHub Actions, HuggingFace

- Built an end-to-end machine learning pipeline using Python, Pandas, and Scikit-learn to predict ship CO_2 emissions
- Designed a pipeline-based architecture (Column Transformer + Pipeline) to ensure consistent preprocessing and avoid serialization issues
- Achieved R^2 score of 0.995 and RMSE of ~1013, indicating high prediction accuracy
- Developed an interactive web app using Streamlit with dynamic UI components (sliders, dropdowns) for real-time predictions
- Deployed the application on Hugging Face Spaces, handling environment and dependency management

- Debugged and resolved production issues including model serialization errors, dependency conflicts, and Git LFS artifact handling
- Implemented a fallback mechanism to retrain model dynamically when deployment artifacts were missing or corrupted
- **Live App :** https://huggingface.co/spaces/rshewatkar/ship_co2-emission-predictor
- **GitHub :** https://github.com/rshewatkar/Ship_Co2_Emission_Prediction

Restaurant Demand / Analytics ML Project

Tech Stack: Scikit-learn, Folium, Streamlit, Docker, GitHub Actions, HuggingFace

- Built end-to-end ML pipeline analyzing 9,551 restaurants across 15+ countries for rating prediction, cuisine classification, and personalized recommendations
- Trained and compared 3 ML models (Linear Regression, Decision Tree, Random Forest) achieving best R^2 score with Random Forest regressor
- Developed content-based restaurant recommendation system using TF-IDF vectorization and Cosine Similarity
- Created interactive geographical analysis with Folium maps showing restaurant density, ratings, and city-wise statistics
- Deployed full Streamlit web application on Hugging Face Spaces
- **Live App :** https://rshewatkar-restaurant-ml-project.hf.space
- **GitHub :** https://github.com/rshewatkar/restaurant-ml-project

Vehicle Price Prediction using Machine Learning

- Developed a regression model to predict used vehicle prices based on features like year, mileage, fuel type, and engine size.
- Explored correlations using heatmaps and visualized key trends in pricing behavior.
- Trained and compared multiple models (Linear Regression, Random Forest, Decision Tree); achieved 85% R^2 on test data.
- Deployed insights to identify influential features and improve model accuracy.

Mobile Phone Price Category Prediction using Machine Learning

- Developed a multi-class classification model to predict mobile phone price categories (low/medium/high/very high) based on key specifications using supervised learning.
- Trained and evaluated five models (Logistic Regression, SVC, KNN, Random Forest, XGBoost), achieving 97.5% accuracy with Logistic Regression.
- Delivered actionable business insights, enabling data-driven pricing strategy for mobile retailers and e-commerce platforms.